

Questions Unit 1

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Questions on Probability

Conceptual Questions

1. Define a sample space and provide an example of a sample space when rolling two dice.
2. What is an event in probability theory? Illustrate with an example.
3. Differentiate between mutually exclusive events and independent events.
4. Explain the concept of conditional probability. Provide a real-world example to demonstrate it.
5. What is Bayes' theorem? Explain its significance with a practical application.

Application-Based Questions

1. A fair coin is tossed twice. What is the probability that at least one head appears?
2. Two dice are rolled. What is the probability that the sum of the dice equals seven?
3. If a card is drawn from a deck of cards, what is the probability that it is a face card given that it is red?
4. A family has two children. Given that one of them is a boy, what is the probability that both children are boys?
5. Suppose that a box contains 3 red balls, 2 blue balls, and 1 green ball. If two balls are drawn at random without replacement, what is the probability that both are red?

True/False Questions

1. The probability of the union of two mutually exclusive events is the sum of their individual probabilities.
2. If two events are independent, knowing the outcome of one affects the probability of the other occurring.
3. The probability of the complement of an event is always 1 minus the probability of the event.
4. Conditional probability applies only when both events involved are mutually exclusive.
5. The inclusion-exclusion principle is used to calculate the probability of the union of multiple events.

Problem-Solving Questions

1. Given two dice, calculate the probability of getting a sum of eight if one of the dice shows a four.
2. An urn contains 4 white balls and 6 black balls. Two balls are drawn one after another without replacement. What is the probability that both balls drawn are black?
3. Three players throw their hats into a pile. What is the probability that none of them picks up their own hat?
4. A blood test is 98% accurate in detecting a disease. However, it also produces a false positive 2% of the time. If 1% of the population has the disease, what is the probability that a person who tested positive actually has the disease?

Advanced Conceptual and Theoretical Questions

1. Explain the inclusion-exclusion principle for calculating the probability of the union of multiple events. Provide an example to illustrate.
2. What is the difference between pairwise independent and jointly independent events? Can events be pairwise independent but not jointly independent? Explain with an example.
3. Derive and explain the formula for the probability of the intersection of two events.
4. What are the properties of a probability measure P on a sample space? How do these properties ensure that the values assigned to events are valid probabilities?

5. Define the complement of an event and show how the probability of the complement is related to the probability of the event itself.

Scenario-Based Questions

1. A biased coin has a probability of 0.6 of landing heads. If the coin is flipped 5 times, what is the probability of getting exactly 3 heads?
2. A deck contains 52 cards. Two cards are drawn without replacement. What is the probability that both cards are aces?
3. A person flips a fair coin repeatedly until a tail appears. What is the probability that the person flips the coin exactly 4 times?
4. In a class of 100 students, 60 are enrolled in Mathematics, 40 in Physics, and 20 in both subjects. A student is chosen at random. What is the probability that the student is enrolled in Mathematics or Physics?
5. A die is rolled. What is the probability that the result is either an even number or greater than 4?

Proof-Based Questions

1. Prove that if two events are mutually exclusive, their intersection is the null event and their joint probability is zero.
2. Prove that if events E and F are independent, then E^c and F are also independent.
3. Given three events A , B , and C , prove that the probability of their union is given by the inclusion-exclusion formula.
4. Prove that for any two events E and F , $P(E \cap F) \leq P(E)$.
5. Prove the law of total probability using the definition of conditional probability.

Problem-Solving and Calculation-Based Questions

1. In a lottery, a person picks 6 numbers out of 49. What is the probability that exactly 3 of the numbers match the winning numbers?
2. A factory produces 95% non-defective items and 5% defective items. If a randomly selected item is found to be defective, what is the probability that it came from Machine A, which produces 30% of the factory's output and has a defect rate of 10%?

3. A family has three children. What is the probability that all three children are boys given that the first child is a boy?
4. In a poker hand of 5 cards dealt from a standard deck, what is the probability of getting a full house (three cards of one rank and two cards of another rank)?
5. In a group of 25 people, what is the probability that at least two people share the same birthday?

Conditional Probability and Bayes' Theorem Applications

1. You are testing for a rare disease that affects 1% of the population. The test is 99% accurate (99% true positive and 1% false positive). If someone tests positive, what is the probability they actually have the disease?
2. There are two urns, one containing 3 white balls and 7 black balls, and the other containing 8 white balls and 2 black balls. An urn is selected at random, and a ball is drawn. If the ball is white, what is the probability that it came from the second urn?
3. A person randomly selects one of three bags. Bag 1 contains 1 red and 4 black balls, Bag 2 contains 2 red and 3 black balls, and Bag 3 contains 3 red and 2 black balls. What is the probability that the person selected Bag 1 given that they drew a red ball?
4. A box contains 4 defective items out of 20. Two items are randomly selected. What is the probability that the second item is defective given that the first item was not defective?
5. You draw a card from a standard deck. If it is a red card, you draw again without replacement. If it is black, you stop. What is the probability that you draw a second card?

Questions on Random Variables

Conceptual Questions

1. Define a random variable. How does it differ from the outcome of an experiment?
2. What is the cumulative distribution function (CDF) of a random variable? List its key properties.
3. Differentiate between discrete and continuous random variables. Provide an example of each.

4. Explain what a probability mass function (PMF) is for a discrete random variable.
5. What is the probability density function (PDF) of a continuous random variable? How is it related to the CDF?

Application-Based Questions

1. A coin is tossed until a head appears. Let X be the number of tosses required. Write the probability mass function (PMF) for X .
2. The sum of the numbers on two fair dice is observed. Define a random variable for this experiment and compute the probability that the sum is 7.
3. Explain the geometric random variable and compute the probability that it takes exactly 5 trials to get the first success in an experiment where the success probability is 0.2.
4. Suppose X is a Poisson random variable with parameter $\lambda = 4$. What is the probability that $X = 2$?
5. Given a normal distribution with mean $\mu = 0$ and variance $\sigma^2 = 4$, find the probability that the random variable X lies between -1 and 1 .

Problem-Solving Questions

1. A binomial random variable is defined by $X \sim \text{Binomial}(n = 4, p = 0.5)$. Compute $P(X = 2)$.
2. For a uniform random variable distributed between 0 and 10, compute $P(2 < X < 8)$.
3. The lifetime of a battery follows an exponential distribution with a rate parameter $\lambda = 0.5$. What is the probability that the battery lasts more than 3 years?
4. Compute the expected value and variance of a geometric random variable with parameter $p = 0.3$.
5. If X and Y are independent random variables with the following joint probability mass function:

$$p(x, y) = \begin{cases} 0.2 & \text{if } x = 1, y = 1 \\ 0.3 & \text{if } x = 1, y = 2 \\ 0.5 & \text{if } x = 2, y = 2 \end{cases}$$

compute $P(X = 1)$ and $P(X + Y = 3)$.

True/False Questions

1. The probability that a continuous random variable takes any specific value is always zero.
2. A Bernoulli random variable can take on only two values: 0 and 1.
3. The Poisson distribution is used to model the number of successes in a fixed number of trials.
4. The expectation of a random variable always equals its median.
5. If two random variables are independent, then their covariance is always zero.

Advanced Questions

1. Explain how Bayes' theorem can be applied to a scenario where you have a geometric distribution.
2. Show how the Poisson distribution can be used to approximate a binomial distribution when n is large and p is small.
3. Prove that the variance of a binomial random variable $X \sim \text{Binomial}(n, p)$ is $np(1 - p)$.
4. Derive the expectation of an exponential random variable with parameter λ .
5. For two jointly distributed continuous random variables X and Y , define and explain the meaning of the joint probability density function (PDF).

Conceptual and Theoretical Questions

1. Define a Bernoulli random variable. How is it related to a Binomial random variable?
2. What is the relationship between the cumulative distribution function (CDF) and the probability density function (PDF) of a continuous random variable?
3. What is the Poisson distribution used for? Give an example where it would be applicable in real life.
4. Explain the memoryless property of the geometric and exponential distributions. Provide an example illustrating this property.
5. Define covariance and correlation. How are they related?

Scenario-Based Questions

1. Consider a random variable X that follows a Poisson distribution with parameter $\lambda = 2$. What is the probability that X is less than or equal to 3?
2. A factory produces defective items with probability 0.05. If 20 items are produced, what is the probability that at most 2 items are defective? Assume the number of defective items follows a binomial distribution.
3. Suppose X is uniformly distributed on the interval $[0, 1]$. Calculate $P(0.3 \leq X \leq 0.7)$.
4. A car manufacturer tests 10 engines and finds that the probability that an engine will last more than 5 years is 0.9. What is the probability that exactly 8 of the engines last more than 5 years?
5. In a normal distribution, approximately what percentage of the data lies within two standard deviations from the mean?

Problem-Solving and Calculation-Based Questions

1. A machine produces items, and each item is defective with probability 0.01. If the machine produces 500 items, use the Poisson approximation to find the probability that there are at least 3 defective items.
2. Calculate the variance of a Poisson random variable with parameter $\lambda = 5$.
3. Let X be a binomial random variable with $n = 10$ and $p = 0.4$. Find the expected value and variance of X .
4. If the lifetime of a certain device follows an exponential distribution with rate parameter $\lambda = 2$, compute the probability that the device will last more than 3 years.
5. Two independent random variables X and Y follow a uniform distribution on the interval $[0, 1]$. What is the probability that $X + Y \leq 1$?

True/False Questions

1. The variance of a random variable is always non-negative.
2. The binomial distribution models the number of failures before the first success in a sequence of Bernoulli trials.
3. If X and Y are independent random variables, then $Cov(X, Y) = 0$.
4. A Poisson random variable can only take integer values.
5. For a normal random variable, the probability that $X = \mu$ (the mean) is greater than 0.

Proof-Based Questions

1. Prove that the expectation of a geometric random variable with parameter p is $\frac{1}{p}$.
2. Show that the cumulative distribution function (CDF) for a uniform random variable on the interval (a, b) is $F(x) = \frac{x-a}{b-a}$ for $a \leq x \leq b$.
3. Derive the formula for the variance of a binomial random variable.
4. Prove that for a Poisson random variable X with parameter λ , the variance is equal to the mean λ .
5. Show that if X is a normal random variable with mean μ and variance σ^2 , then the transformed variable $Y = \frac{X-\mu}{\sigma}$ is a standard normal random variable.

Advanced Questions

1. Explain how the law of total probability can be applied to calculate the expectation of a random variable given a conditional distribution.
2. Derive the formula for the expectation of a Poisson random variable using the moment generating function.
3. A continuous random variable X has a PDF given by $f(x) = 4x^3$ for $0 \leq x \leq 1$. Compute the mean and variance of X .
4. Given two random variables X and Y with joint distribution $p(x, y)$, show how to compute the conditional expectation $E[X | Y = y]$.
5. Define jointly distributed random variables. How would you compute the covariance of two random variables given their joint probability distribution?